Brookhaven Spreads Out

AP Human Geography · Unit 6 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

Brookhaven is an invented U.S. metropolitan area of about one million people. Like many American metros, its population has shifted outward since 1970, while a few old neighborhoods near downtown have recently become popular, and expensive, again.

Brookhaven Metropolitan Area: Population, Housing, and Commuting by Zone, 1970 and 2020

Zone	Population 1970	Population 2020	Median home price 2020	Percent of workers who drive alone to work	Percent of residential land zoned single-family only
Central city (inside the old core)	520,000	445,000	$395,000	55%	25%
Inner suburbs	310,000	325,000	$265,000	78%	62%
Outer suburbs / exurbs	95,000	300,000	$380,000	90%	85%
Metro total	925,000	1,070,000	$340,000	72%	71%

Note: The central city's population fell to 388,000 in 2000, then grew again after 2010, mostly in two renovated neighborhoods near downtown (Millside and the Canal District), where the 2020 median home price was $610,000. Brookhaven and all of its data are invented for practice. Class directions: label each part, and write Describe and Explain parts as T1- (your reason in general words), E2- (the specific proof), A3- (why it matters).

A. Identify the process shown by the growth of the outer suburbs and exurbs between 1970 and 2020. IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Suburbanization or urban sprawl (also accept decentralization, exurbanization, outward growth)

MODEL ANSWER

Suburbanization, or urban sprawl: population spreading outward from the central city into low-density outer suburbs.

Common mistake: “Urbanization.” Urbanization means a bigger share of people living in cities overall. This table shows the center LOSING people while the far edge triples. That specific outward movement is suburbanization (sprawl).

Why the model works: It names the specific process (suburbanization/sprawl) and ties it to what the table shows: outward, low-density growth at the edge.

B. Using the table, describe the relationship between distance from the center of Brookhaven and the percent of workers who drive alone to work. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Uses the table's numbers (55%, 78%, 90%)
- States the direction of the relationship: farther out means a higher drive-alone share (or closer in means lower)

MODEL ANSWER

T1- The farther a zone is from the center of Brookhaven, the higher the percent of workers who drive alone to work.
E2- In the central city only 55% drive alone, in the inner suburbs it rises to 78%, and in the outer suburbs and exurbs it reaches 90%.
A3- This fits because the outer zones are low-density and built around the car: homes are far from jobs and shops and there is little bus or rail service, while the center has transit, sidewalks, and jobs close together, so more people can walk, bike, or ride.

Common mistake: “Most people in Brookhaven drive to work alone.” True, but it's one fact about the whole metro. The question asks how the drive-alone share CHANGES with distance: compare the three zones and quote their numbers.

Why the model works: It states the direction (farther out = more driving alone), quotes all three numbers, and explains the pattern with density and transit.

C. Explain ONE cause of the growth of Brookhaven's outer suburbs shown in the table. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a cause: cars/highways, cheaper land, government mortgage policy (FHA/VA loans), white flight, schools, more space, lower taxes or crime
- Explains how that cause produced growth at the edge (made moving out possible, cheaper, or attractive)

MODEL ANSWER

T1- One cause of the outer suburbs' growth is the spread of cars and highways.
E2- After the Interstate Highway System began in 1956, most families owned a car, so a worker could live 20 or 30 miles from downtown on cheap land and still commute to a job in the city.
A3- Because developers could build bigger, newer houses on inexpensive land at the edge and highways made the daily trip possible, families moved outward, and Brookhaven's outer suburbs tripled from 95,000 to 300,000 people while the center lost population.

Common mistake: “People wanted to live in the suburbs.” That's a wish, not a cause. What made it POSSIBLE to live far from downtown? Name something specific (highways and cars, cheap land, mortgages) and explain how it led people to move.

Why the model works: It names a specific cause (cars and highways), gives dated proof (the 1956 Interstate system), and explains the chain all the way to the table's numbers.

D. Explain how the zoning data in the table could contribute to high housing prices in Brookhaven. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Identifies the constraint: single-family-only zoning blocks apartments and denser housing, limiting the number of homes (supply)
- Explains the link to price: limited supply plus demand leads to competition and higher prices

MODEL ANSWER

T1- Single-family-only zoning limits how many homes can be built, which raises prices.
E2- In Brookhaven, 85% of the outer suburbs' residential land and 71% of the metro's allows only one detached house per lot, so apartments, duplexes, and townhouses that could fit many more families are illegal on most of the land.
A3- Because the metro added about 145,000 people but zoning keeps the supply of homes small, buyers compete for the few houses available and bid prices up, which helps explain the $380,000 median in the outer suburbs.

Common mistake: “Rich people live in the outer suburbs, so houses cost more there.” That describes who lives there, not how ZONING works. The question is about the single-family-only percentage: explain how limiting what can be built limits supply and raises prices.

Why the model works: It names the constraint (single-family-only zoning limits supply), quotes the table (85%, 71%), and walks through supply, demand, and price.

E. Explain how the recent growth and high home prices in some of Brookhaven's central-city neighborhoods could be an example of gentrification, including ONE effect on long-time residents. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes gentrification: wealthier or younger people moving into and renovating older central neighborhoods, raising values
- Gives one effect on long-time residents: displacement, unaffordable rent or taxes, loss of community or culture

MODEL ANSWER

T1- The comeback of a few central-city neighborhoods looks like gentrification: wealthier newcomers moving into older, lower-income areas near downtown.
E2- Brookhaven's central city shrank from 520,000 to 388,000 by 2000, but since 2010 it has grown again mostly in the renovated Millside and Canal districts, where the median home price is $610,000, far above the inner suburbs' $265,000.
A3- As investors and young professionals renovate old homes, rents and property taxes rise, so long-time lower-income residents, especially renters, can be displaced because they can no longer afford to stay in their own neighborhood.

Common mistake: “The city is growing again because it is popular now.” 'Popular' isn't a process. Say WHO is moving in and WHY (wealthier people buying and renovating cheap old housing near downtown), then give the effect on the people already there.

Why the model works: It defines gentrification, anchors it in the note ($610,000 Millside and Canal districts, growth since 2010), and gives a clear effect: displacement by rising rents and taxes.

F. Describe ONE smart-growth or New Urbanism policy Brookhaven could adopt to reduce sprawl, and how it would work. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a smart-growth or New Urbanism policy: urban growth boundary, greenbelt, transit-oriented development, mixed-use zoning, infill or brownfield redevelopment, denser housing, walkable design
- Describes how it works to reduce sprawl (keeps growth inside, near transit, denser, less driving, protects land at the edge)

MODEL ANSWER

T1- Brookhaven could use transit-oriented development (TOD) to pull growth inward instead of outward.
E2- The city would rezone the land within about a half mile of its rail and bus stations for apartments, townhouses, shops, and offices, the way Arlington, Virginia built dense, walkable neighborhoods around its Metro stations.
A3- Because thousands of new homes would be built inside the existing metro next to transit, fewer families would need a new house on farmland at the edge, and more workers could live where the drive-alone share is 55% rather than 90%, which reduces sprawl and traffic.

Common mistake: “Build more roads so people can get to work faster.” More roads is the opposite of smart growth: wider highways make living even farther out easier, which causes MORE sprawl. Pick a policy that keeps growth inside the city.

Why the model works: It names a recognized policy (TOD), describes exactly what changes (rezoning near stations) with a real example, and connects it to less sprawl and less driving.

G. Explain ONE limitation of using data for Brookhaven's three large zones, or the metro total, to understand its housing or transportation problems. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a limitation: zone-level or metro-level averages hide variation within zones or neighborhoods, or city/suburb boundaries are defined differently, or rent and income data are missing
- Explains why it misleads or what data would be needed instead

MODEL ANSWER

T1- One limitation is scale: each row averages hundreds of thousands of people, so it hides big differences between neighborhoods.

E2- The central city's median home price is listed as $395,000, but the note says the renovated Millside and Canal districts are at $610,000, which means other central neighborhoods must be far cheaper; likewise the 55% drive-alone figure lumps transit-rich downtown together with car-dependent parts of the city.

A3- Because the zone averages blur these differences, a planner could not tell where housing is actually unaffordable or where buses are missing, and would need neighborhood or census-tract data to target the real problem.

Common mistake: “The data is from 2020, so it is out of date.” 2020 is the most recent census year, and the question is about the SIZE of the areas, not the date. Explain what three giant zones hide, such as differences between neighborhoods.

Why the model works: It names the limitation (scale and averages), proves it with the $610,000 vs. $395,000 gap, and explains how it could mislead a planner.

Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.